PROJECT DOCUMENTATION

Stakeholder Analysis, User Stories & Use Cases

Supply Chain Performance Analysis

Shipment Delays, Cost Efficiency, Inventory Availability, and the Impact of Shortages on Patient Cases

1. Purpose & Scope

The proposed scenario assumes that a central health authority monitors supply chain performance across hospitals.

Project Objectives

- Monitor shipment delays and delivery-cycle performance.
- Assess supplier performance and logistics cost efficiency.
- Identify inventory availability issues and unmet demand.
- Explore descriptive patterns between treatment status and patient-case indicators.

2. Stakeholder Analysis

Stakeholders are represented as role-based personas rather than named individuals. Their needs are linked to the dashboard areas and decisions they may support.

ID	Stakeholder	Primary Responsibility	Information Needs	Relevant Dashboard Area
ST-01	Central Executive Management	Strategic oversight and prioritization	Executive KPIs, shipment performance trends, key risks, improvement opportunities	Executive Overview; Case Impact & Decision Quality
ST-02	Central Supply Chain Director	Monitor delivery and operational performance	Late rates, delay distributions, lead time, country and mode comparisons	Executive Overview; Delivery & Cycle Time
ST-03	Procurement & Supplier Performance Manager	Review supplier performance and logistics costs	Supplier late rates, delayed shipment value, freight cost, cost-to-value ratios	Suppliers & Cost
ST-04	Hospital Inventory / Pharmacy Manager	Monitor stock availability and operational replenishment needs	Stockout rates by hospital/item, criticality, estimated unmet-demand value	Purchase Orders & Inventory
ST-05	Quality Team	Review patterns in recorded patient-case indicators	Case status, treatment delay, recorded outcomes and complications	Case Impact & Decision Quality
ST-06	Chief medical officer (CMO)	Understand case mix and review potential service-delivery concerns	Case counts, treatment-delay distribution, diagnosis/hospital patterns	Case Impact & Decision Quality

Stakeholder Engagement Considerations

- Executive users need concise indicators, trends, and clearly prioritized findings.
- Operational users need filters and enough detail to investigate a specific supplier, shipment, hospital, or item.
- Medical and quality users need transparent definitions and cautious interpretation of patient-related comparisons.

3. Analytical Questions by Stakeholder

The following 21 analytical questions are assigned to the six original stakeholders identified in the Stakeholder Analysis. The question wording is retained, with the stakeholder labels supplied in the original question list removed.

ST-01 — Central Executive Management

Q01: I want to know how many shipments were recorded, their total value, and how the value splits by year and by product group, so that the size and composition of the shipment portfolio is known.

Q02: I want to know the headline hospital figures: purchase orders, inventory days and cases, with PO On-Time Rate, Stockout Rate and Unmet Demand EGP, so that the hospital layer can be read next to the shipments on one page.

Q03: I want to know how complete the key fields of both families are after cleaning, so that the reliability of every other page is visible.

Q21: I want to know the top improvement opportunities supported by the numbers and how stable the results are after cleaning, so that every recommendation is linked to a measure, a sample size and stated limits.

ST-02 — Central Supply Chain Director

Q04: I want to know how many shipments were delivered on schedule, earlier or later than scheduled, so that delivery performance is stated with its denominator.

Q05: I want to know the distribution of lead time from PO to delivery recorded, and the median lead time by country, so that cycle time is reported with sample size, for countries with at least 5 shipments.

Q06: I want to know whether customs clearance days relate to lead time, so that the share of cycle time spent in customs is understood without implying causation.

Q07: I want to know how much shelf life remained at delivery, which shipments arrived already expired, and the receiving inspection results, so that expiry risk at delivery is identified from the records.

ST-03 — Procurement & Supplier Performance Manager

Q08: I want to know which vendors account for the largest share of shipment value, so that vendor concentration is known; vendors with fewer than 3 shipments are grouped as Other.

Q09: I want to know how value and unit price differ by product group and molecule or test type, so that price and value drivers are compared within the same molecule.

Q10: I want to know the total freight and insurance cost and the Freight/Value ratio by country and by product group, so that cost efficiency is computed from valid numeric records only.

Q11: I want to know whether freight cost grows with weight, the Freight per Kg, and where the outliers are, so that outliers are flagged for review instead of being deleted.

ST-04 — Hospital Inventory / Pharmacy Manager

Q12: I want to know each supplier's actual on-time rate against its historical rate, and its PO fill ratio, so that the difference is read with the caveat that period and measurement may differ.

Q13: I want to know how many days received purchase orders are late and how orders split by status, so that the PO delay distribution and the status mix are known.

Q14: I want to know how actual order-to-receipt time compares with the item baseline lead time, so that categories where actual lead time exceeds the baseline are identified.

Q15: I want to know where the Stockout Rate is concentrated by hospital and criticality, and how it moves week by week, so that the rate and the count are shown, not the quantity only.

Q16: I want to know the estimated value in EGP of unmet demand by item and of wastage by item category and cold-chain flag, so that values are described as estimates and not as realized losses.

ST-05 — Quality Team

Q17: I want to know how many cases fall in each treatment status and how treatment delay hours are distributed, so that denominators and missing data are visible.

Q19: I want to know how each item's Stockout Rate compares with the share of its cases delayed because stock was unavailable, so that the descriptive link between item stockouts and delayed treatment is visible.

ST-06 — Chief medical officer (CMO)

Q18: I want to know how outcome, ICU admission, complication and length of stay differ by treatment status, and the death share within each severity level, so that the comparison is descriptive, with group sizes shown and no causal claim.

Q20: I want to know which diagnoses and hospitals have higher shares of shortage-type cases, so that affected diagnoses and hospitals are named, above a minimum group size.

4. Sample Use Cases

Use cases describe a practical interaction with the dashboard, including the intended outcome and relevant constraints. These examples are deliberately limited in scope for an initial review.

UC-01 | Monitor Overall Shipment Performance

Field	Description
Primary Actor	Central Executive Manager
Dashboard Area	Executive Overview
Goal	Assess overall shipment performance and identify trends that require attention.
Preconditions	Relevant shipment records have been cleaned; KPI definitions and date-validity rules have been agreed.
Trigger	The stakeholder opens the executive dashboard for a selected reporting period.
Main Flow	1. Review shipment volume and late-delivery rate. 2. Compare performance across months or years. 3. Review average, median, and P90 delay indicators where available. 4. Filter the view to investigate a period or category of interest. 5. Identify trends that warrant operational follow-up.
Expected Outcome	A concise overview of shipment performance and a shortlist of areas for further investigation.
Exceptions / Limitations	Invalid date records are excluded according to the defined KPI rules. Sample size and denominator should remain visible when interpreting rates.

UC-02 | Review Supplier Performance and Cost

Field	Description
Primary Actor	Procurement & Supplier Performance Manager
Dashboard Area	Suppliers & Cost
Goal	Identify suppliers or logistics segments that may require a closer review.
Preconditions	Supplier identities and relevant cost fields are available and validated; minimum-volume rules are documented.
Trigger	The manager opens supplier performance results for a selected period.
Main Flow	1. Compare supplier late rates and shipment counts. 2. Review the value associated with delayed shipments. 3. Compare freight cost and freight-to-shipment-value ratios by mode. 4. Inspect high-performing or high-risk segments before drawing conclusions.
Expected Outcome	A documented shortlist of suppliers or segments for further operational review.
Exceptions / Limitations	Rates should not be ranked without considering sample size. Freight/value is calculated from aggregated valid totals, not by averaging row-level ratios. Association does not prove causation.

UC-03 | Identify Inventory Availability Issues

Field	Description
Primary Actor	Hospital Inventory / Pharmacy Manager
Dashboard Area	Purchase Orders & Inventory
Goal	Identify where recorded unmet demand may indicate stock availability issues.
Preconditions	Valid inventory observations, hospital/item keys, and item criticality information are available.
Trigger	The manager selects a hospital, item, criticality level, or reporting period.
Main Flow	1. Review stockout rate and valid observation count. 2. Compare hospitals or items. 3. Identify critical items with repeated unmet demand. 4. Review estimated unmet-demand value where valid unit-cost data exists. 5. Flag selected items for operational follow-up.
Expected Outcome	A prioritized view of potential stock availability issues for further review.
Exceptions / Limitations	Stockout rate uses valid hospital-item-day records as the denominator. Estimated unmet-demand value is an estimate, not confirmed financial loss.

5. Requirements Traceability

The project should maintain a clear line of traceability so that every key visual supports a defined business question and user need.

Stakeholder → User Story → Analytical Question → KPI / Visual → Use Case

Traceability Element	Review Question
Stakeholder	Who needs this information and why?
User Story	What does the user need to achieve?
Analytical Question	Which project question addresses that need?
KPI / Visual	What measure or visual provides the evidence?
Use Case	How will the user interact with the dashboard and use the result?